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Mono-Crystalline Silicon Solar Cell Optimization and Modeling

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Solar cell manufacturers are under constant pressure to improve cell efficiency, for product differentiation as well as cost per watt reduction. For conventional mono-crystalline silicon cells, one appealing strategy is to introduce point contacts on the rear surface instead of full surface contact.

With point contacts on the rear surface of a solar cell, several competing physical mechanisms determine its performance. On the one hand, different optical reflectivity and surface recombination rates for the silicon-aluminum interface and passivated silicon-nitride interface suggest that reducing rear contact area would boost cell efficiency. On the other hand, current crowding, contact resistance, and bulk recombination will contribute to cell performance degradation with shrinking rear contact area. Furthermore, the trade-off between these factors will be affected by any change in doping, silicon quality and cell size. Therefore, there is a large optimization space to find the best solar cell design.

We perform optimization of solar cell design using three-dimensional simulation with Sentaurus TCAD tools.

Figure 1 illustrates various geometry parameters to define the solar cell structure and simulation domain boundaries, and Fig. 2 shows definition of p-n junctions.

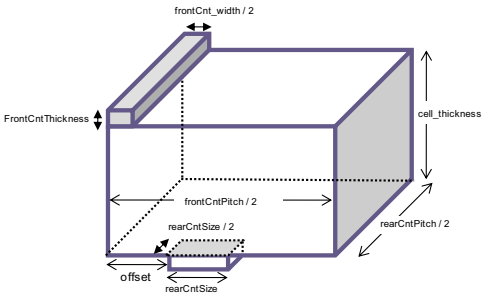

Fig. 1. Definition of structure and simulation domain.

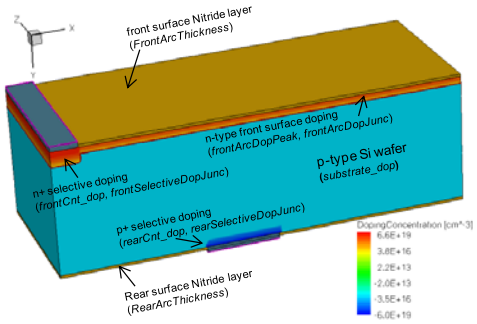

Fig. 2. Doping profiles with blanket n-type top surface doping and local n+ and p+ junctions around the contacts.

For each structure configuration, the simulation work flow starts from processing the optical carrier generation profile that accounts for surface texture and different optical reflectivity at rear surfaces with or without the contact. It is followed by using Sentaurus Structure Editor to create a 3D structure with the processed optical profiles. Then, Sentaurus Device is used to perform electrical analysis of solar cell. In the last step, post-processing of the calculated current-voltage curves is done, with extraction of the key parameters like short-circuit current, maximum power, and overall efficiency. Figure 3 shows the optical generation profiles in a solar cell after the Sentaurus Structure Editor step.

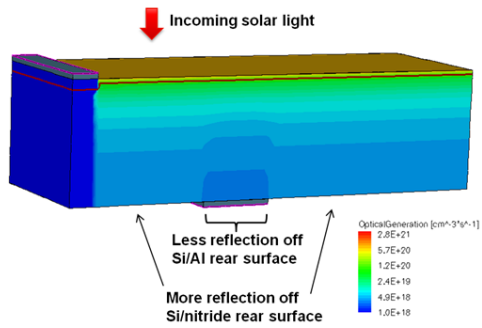

Fig. 3. Optical generation profiles in a solar cell.

Sweeping the sizes of various parts of the solar cell, or depth and doping level of the junctions, or physical properties of silicon and its interfaces, their impact on solar cell performance can be determined.

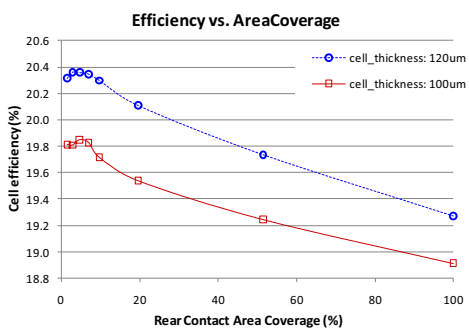

Fig. 4. Efficiency of a solar cell as a function of rear contact area.

Figure 4 demonstrates typical design trade-off, with cell efficiency shown as a function of the rear contact area coverage. The non-monotonic trend is a result of multiple competing physical mechanisms and points towards an optimal design around 4% rear contact area. Comparing three-dimensional analysis to the simplified two-dimensional and one-dimensional cases, we see significant differences in the solar cell behavior that make it necessary to perform a true three-dimensional analysis.

For a different set of p-n junctions, silicon and interface lifetime properties, or contact pitches, the optimum design will be somewhat different, and can be found using the proposed modeling methodology.